Latest version: `https://vuphan314.github.io`
This version: 2020/04/09

Vu H. N. Phan

CV

vuphan314

Education

2017/08–present **Doctor of Philosophy**, *Computer Science*, Rice University, Houston TX.
- ***Advisor: Moshe Vardi***
- Research: Computational Logic

2017/08–2019/12 **Master of Science**, *Computer Science*, Rice University, Houston TX.
- ***Advisor: Moshe Vardi***
- Research: Computational Logic
- Thesis: Weighted Model Counting with Algebraic Decision Diagrams

2014/08–2017/08 **Dual Bachelor of Science**, *Computer Science & Math*, Texas Tech University, Lubbock TX.
- ***Advisor: Michael Gelfond***
- Research: Logic Programming

2013/01–2014/08 **Associate of Science**, Collin College, Plano TX.

Research

Published

2018–2020 **ADDMC: Weighted Model Counting with Algebraic Decision Diagrams**.
- Employed dynamic programming to develop the exact model counting framework `ADDMC` with Jeffrey Dudek and Moshe Vardi.
- Presented our paper at the AAAI conference in New York City on 2020/02/11.

Supervisor: Moshe Vardi (Computer Science Department, Rice University).

2016–2018 **Syntactic Conditions for Antichain Property in `CR-Prolog`**.
- Found syntactic conditions guaranteeing when `Consistency Restoring Prolog` programs have antichain property (no answer set is a proper subset of another).
- Presented my paper at the ASPOCP workshop (affiliated with ICLP, part of FLoC) in Oxford on 2018/07/18.

Supervisor: Michael Gelfond (Computer Science Department, Texas Tech University).

Unpublished

2013–2017 **Biquasiarithmetic Mean Comparability**.
- Generalized a theorem on inequalities involving quasiarithmetic means.
- Resubmitted my paper to the Rose-Hulman Undergraduate Math Journal in 2017 (previously submitted in 2013 & 2014).
- Presented my paper at the Texas Undergraduate Math Conference in San Antonio on 2013/10/26.

Supervisor: Alan Graves (Math Department, Collin College).

Experience

Software Developing

2016–2017 `LED`, *Literate Programming System*.

- Implemented the `Language of Effective Definitions` by automating its translation into the functional programming language `SequenceL`.
- Implemented an `LED-to-TeX` translator.

Supervisor: Nelson Rushton (Computer Science Department, Texas Tech University).

2015–2016 `L`, *Logic Programming Engine*.

- Implemented the language `L` by automating its translation into `Answer Set Prolog`.

Supervisor: Nelson Rushton (Computer Science Department, Texas Tech University).

Tutoring

2015, 2017 **Computer Science, Math, Physics, Chemistry**.
Supervisor: Patrick Bohn (Learning Center, Texas Tech University)

2014 **Math, Physics, Chemistry**.
Supervisor: Dennis Keeton (Math Lab, Collin College)

Scholarships

2017–present **Ph.D. Student Tuition & Stipend**, *Computer Science Department*, Rice University.
2016–2017 **General Scholarship**, *Financial Aid Office*, Texas Tech University.
2015–2017 **Academic Achievement Scholarship**, *College of Arts & Sciences*, Texas Tech University.
2015–2017 **Undergraduate Achievement Scholarship**, *Math Department*, Texas Tech University.
2015–2016 **Undergraduate Research Scholarship**, *Math Department*, Texas Tech University.
2014–2016 **Proven Achiever Scholarship**, *Financial Aid Office*, Texas Tech University.
2013–2014 **Excellence Scholarship**, *Math Department*, Collin College.

Courses

2020 Spring **Statistical Machine Learning**.
2019 Fall **Computer Systems Architecture**.
2019 Spring **Compiler Construction; Multi-Core Computing**.
2018 Fall **Artificial Intelligence**.
2018 Spring **Automated Program Verification; Principles of Programming Languages**.
2017 Fall **Advanced Logic in Computer Science; Functional Programming**.

Computer Skills

- Programming languages:
  - Procedural: `Python`, `C++`, `Java`
  - Functional: `Scala`, `Scheme`
  - Logic: `SPARC`, `Clingo`, `DLV`
  - Data: `SQL`
  - Formatting: `TeX`, `HTML`
- Tools: Git, Linux/Windows shells